CHM2045C: Final Exam Comprehensive Study Guide

Chemistry Study Guide Draft

Spring 2026

Chapter 1: Foundations: The Chemical Toolbox

This chapter establishes the fundamental quantitative and qualitative tools required for chemical analysis, focusing on measurement, classification, and naming.

Density and Volume Displacement

Density is an intrinsic property relating mass and volume. For irregular solids, volume is determined by the displacement of a liquid.

Key Formula *

$$d = \frac{m}{V} \quad \text{and} \quad V_{\text{object}} = V_{\text{final}} - V_{\text{initial}}$$

1. Identify given mass (g) and volume (mL or cm^3).
2. Use displacement to find V if necessary (Archimedes' Principle).
3. Solve for the unknown variable using algebraic rearrangement.

Chemical Nomenclature

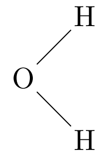
- **Ionic Compounds**: Name the cation followed by the anion. Use Roman numerals for transition metals with variable oxidation states (e.g., $FeCl_3$ is iron(III) chloride).
- **Acids**: Binary acids (H + element) use the prefix *hydro-* and suffix *-ic* (e.g., HCl(aq) is hydrochloric acid). Oxyacids use *-ic* for *-ate* ions and *-ous* for *-ite* ions.
- **Hydrates**: Append the number of water molecules using Greek prefixes (e.g., $CuSO_4 \cdot 5\,H_2O$ is copper(II) sulfate pentahydrate).

Classification of Matter

Matter is classified by its composition and uniformity at the particulate level.

- **Elements**: Pure substances consisting of one type of atom (e.g., O_2).
- **Compounds**: Pure substances with fixed ratios of different atoms (e.g., H_2O).
- **Mixtures**: Physical blends of substances. **Homogeneous** (uniform) or **Heterogeneous** (non-uniform).

Particulate Representation of a Water Molecule (H_2O):



Metric System, Units, and Temperature

Mastery of prefixes is essential for dimensional analysis and unit comparison.

Prefix	Nano (n)	Micro (μ)	Milli (m)	Kilo (k)
Factor	10^{-9}	10^{-6}	10^{-3}	10^{3}

Temperature Conversion (Provided)

$$T_{\mathrm{K}} = T_{^\circ\mathrm{C}} + 273.15$$

Note: Kelvin is mandatory for all gas law and thermodynamic calculations. Absolute zero is 0 K.

Chapter 2: Atomic Structure

Understanding the atom involves identifying its constituent subatomic particles and their contribution to the element's identity and mass.

Nuclide Notation and Subatomic Particles

The identity of an atom is defined by its atomic number (Z), while its mass is determined by the sum of protons and neutrons.

Key Formula *

$$A = Z + N \quad \text{and} \quad \text{Charge} = \text{Protons} - \text{Electrons}$$

In the symbol $^{\text{A}}_{\text{Z}}\text{X}^{\text{charge}}$:

- **Protons**: Equal to Z (Atomic Number). Defines the element.
- **Neutrons**: Calculated as $A - Z$ (Mass Number - Atomic Number).
- **Electrons**: Calculated as $Z - (\text{Charge})$. In neutral atoms, $e^- = p^+$.

Average Atomic Mass

The mass reported on the periodic table is a weighted average of all naturally occurring isotopes based on their abundance.

Key Formula *

$$\text{Avg. Atomic Mass} = \sum_i (\text{Mass of Isotope}_i \times \text{Fractional Abundance}_i)$$

Technique: Convert percentages to decimals (e.g., 75% → 0.75) to find the "fractional abundance" before multiplying by isotopic mass.

Isotopes vs. Ions

- **Isotopes**: Atoms of the same element (same Z) with different numbers of neutrons (N), resulting in different mass numbers (A).
- **Ions**: Atoms that have gained or lost electrons, resulting in a net positive (cation) or negative (anion) charge.

Chapter 3: Electronic Structure and Periodic Trends

Electromagnetic Radiation and the Bohr Model

Light behaves as both a wave and a particle (photon). The energy of an electron in an atom is quantized.

Light and Photon Energy *

$$c = \lambda\nu \quad \text{and} \quad E = h\nu = \frac{hc}{\lambda}$$

$$\Delta E = -2.18 \times 10^{-18}\text{J}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

Note: h and c are usually provided, but the relationships between E, λ, and ν must be mastered.

Quantum Numbers

Four quantum numbers describe the unique quantum state (or "address") of an electron:

1. **Principal (n)**: Energy level $(1, 2, 3\ldots)$.
2. **Angular Momentum (l)**: Orbital shape (0 to $n-1$). $s = 0, p = 1, d = 2, f = 3$.
3. **Magnetic (m_l)**: Orbital orientation ($-l$ to $+l$).
4. **Spin (m_s)**: Direction of electron spin ($+1/2$ or $-1/2$).

Electron Configurations

Follow three core principles to populate orbitals in the ground state:

- **Aufbau Principle**: Electrons fill the lowest energy orbitals first $(1s, 2s, 2p\ldots)$.
- **Pauli Exclusion Principle**: No two electrons can have the same four quantum numbers (max 2 e^- per orbital with opposite spins).
- **Hund's Rule**: Electrons fill degenerate orbitals singly with parallel spins before pairing.

Noble Gas Shorthand: Use the preceding noble gas to represent core electrons (e.g., Sn is $[\text{Kr}]\,5\text{s}^2\,4\text{d}^{10}\,5\text{p}^2$).

Periodic Trends: Ionization Energy (IE)

Ionization Energy is the energy required to remove an electron from a gaseous atom.

- **Across a Period**: IE generally increases due to increasing effective nuclear charge (Z_{eff}).
- **Down a Group**: IE generally decreases due to increasing atomic radius and electronic shielding.

Chapter 4: Formula Calculations and the Mole

Core Concepts

The mole is the fundamental unit in chemistry for measuring the amount of a substance. It provides a bridge between the microscopic world of atoms and molecules and the macroscopic world of grams and liters.

- **Avogadro's Number**: 1 mole = 6.022×10^{23} representative particles.
- **Molar Mass**: The mass in grams of one mole of a substance (g/mol).
- **Percent Composition**: The percentage by mass of each element in a compound.

Avogadro Constant (Provided)

$$N_A = 6.022 \times 10^{23} 1/\text{mol}$$

Mole Calculations *

$$n = \frac{m}{MM} \quad \text{and} \quad N = n \times N_A$$

Problem-Solving Skills

1. **Mass-to-Particle Conversions**:
   - Mass (g) $\xrightarrow{\div MM}$ Moles $\xrightarrow{\times N_A}$ Particles
2. **Calculating Empirical Formulas**:
   a) Assume a 100 g sample (convert % directly to grams).
   b) Convert grams of each element to moles using atomic masses.
   c) Divide all mole values by the smallest mole value obtained.
   d) If necessary, multiply by a small integer to reach whole numbers.
3. **Percent Mass Composition**:

   Key Formula *

$$\text{Mass \% of element} = \left(\frac{\text{mass of element in 1 mol compound}}{\text{molar mass of compound}}\right) \times 100\%$$

Chapter 5: Stoichiometry

Core Concepts

Stoichiometry involves using the relationships between reactants and products in a balanced chemical equation to determine quantitative data.

- **Mole Ratio**: Derived from the coefficients of a balanced equation; it is the "bridge" that allows conversion between different substances.
- **Limiting Reactant**: The reactant that is completely consumed first, limiting the amount of product formed.
- **Theoretical Yield**: The maximum amount of product that can be produced from a given amount of limiting reactant.

Reaction Efficiency *

$$\text{Percent Yield} = \left(\frac{\text{Actual Yield}}{\text{Theoretical Yield}}\right) \times 100\%$$

Problem-Solving Skills

1. **Mass-to-Mass Stoichiometry**:
   - $\text{Mass A} \xrightarrow{\div MM_A} \text{Moles A} \xrightarrow{\text{Mole Ratio}} \text{Moles B} \xrightarrow{\times MM_B} \text{Mass B}$
2. **Identifying the Limiting Reactant**:
   a) Convert the given amounts of all reactants to moles.
   b) Divide the moles of each reactant by its coefficient in the balanced equation.
   c) The reactant with the smallest resulting value is the **limiting reactant**.
3. **Calculating Theoretical Yield**:
   - Use the moles of the limiting reactant and the mole ratio from the balanced equation to find the moles (and then mass) of the desired product.

Chapter 6: Solutions and Aqueous Reactions

Core Concepts

Most chemical reactions occur in solution. Understanding concentration and the behavior of ions in water is critical.

- **Molarity (M)**: A measure of concentration defined as moles of solute per liter of solution (mol/L).
- **Electrolytes**: Substances that dissociate into ions in water (Strong = complete dissociation, Weak = partial).
- **Precipitation**: A reaction where two aqueous solutions form an insoluble solid product.
- **Oxidation States**: Bookkeeping for electrons to track redox processes.

Concentration and Dilution *

$$M = \frac{n}{V} \quad \text{and} \quad M_1V_1 = M_2V_2$$

Solubility Reference

Table 1: Solubility Guidelines for Ionic Compounds *

Always Soluble	NO_3^-, $C_2H_3O_2^-$, Group 1 (Li^+, Na^+, K^+), NH_4^+
Generally Soluble	Cl^-, Br^-, I^- (except Ag^+, Pb^{2+}, Hg_2^{2+}) SO_4^{2-} (except Ba^{2+}, Sr^{2+}, Ca^{2+}, Pb^{2+}, Ag^+)
Generally Insoluble	S^{2-}, OH^- (except Group 1, NH_4^+; Ca^{2+}, Sr^{2+}, Ba^{2+} are soluble) CO_3^{2-}, PO_4^{3-} (except Group 1 and NH_4^+)

Problem-Solving Skills

1. **Writing Net Ionic Equations**:
   a) Write the balanced molecular equation.
   b) Use solubility rules to identify precipitates ((s)) and aqueous species ((aq)).
   c) Write the complete ionic equation (dissociate all strong electrolytes).
   d) Cancel **spectator ions** (ions that appear identical on both sides).
2. **Assigning Oxidation States**:
   - Elements in standard state = 0; Monoatomic ions = charge.
   - F = -1; O = -2 (usually); H = +1 (with non-metals).
   - The sum of oxidation states must equal the overall charge of the molecule/ion.

3. **Solution Stoichiometry & Titration**:

- Use volume and molarity to find moles ($n = M \times V$).
- Apply stoichiometry from the balanced neutralization equation (e.g., $HCl + NaOH \longrightarrow NaCl + H_2O$).
- At the equivalence point: moles H^+ = moles OH^-.

Chapter 7: Heat and Enthalpy

Core Concepts

- **Calorimetry**: The measurement of heat flow into or out of a system during a chemical or physical process.
- **Conservation of Energy**: In an insulated system, heat lost by one component is gained by another: $-q_{\text{lost}} = q_{\text{gained}}$.
- **Standard Enthalpy of Formation (ΔH_f°)**: The change in enthalpy when 1 mole of a substance is formed from its elements in their standard states. Note: $\Delta H_f^\circ = 0$ for any element in its most stable form (e.g., O_2(g), C(graphite)).
- **Hess's Law**: If a reaction is carried out in a series of steps, ΔH for the overall reaction equals the sum of the enthalpy changes for the individual steps.

Calorimetry

Specific Heat Equation (Provided)

$$q = mc_s\Delta T$$

Reaction Enthalpy Calculations

Enthalpy Calculations *

From Heats of Formation:

$$\Delta H^\circ_{\text{rxn}} = \sum n\Delta H_f^\circ(\text{products}) - \sum m\Delta H_f^\circ(\text{reactants})$$

From Bond Energies:

$$\Delta H^\circ_{\text{rxn}} = \sum(\text{bonds broken}) - \sum(\text{bonds formed})$$

Crucial Distinction: Formation uses (Products - Reactants), while Bond Energy uses (Reactants/Broken - Products/Formed).

Problem-Solving Skills

1. **Calculating Reaction Enthalpy**:
   - Identify ΔH_f° for all species. Pure elements in standard states are $0\,\text{kJ/mol}$.
   - Multiply each value by its stoichiometric coefficient.
   - Apply: Products − Reactants.
2. **Applying Hess's Law**:
   - If you reverse a reaction, flip the sign of ΔH.

- If you multiply coefficients by a factor n, multiply ΔH by n.
- Sum the modified ΔH values to find the target enthalpy.

Chapter 8: Structure and Bonding

Core Concepts

- **Formal Charge**: Used to determine the most plausible Lewis structure. The best structure minimizes formal charges and places negative charges on the most electronegative atoms.
- **Lattice Energy**: The energy required to separate 1 mole of an ionic solid into gaseous ions. It increases with increasing ion charge and decreasing ion size.
- **VSEPR Theory**: Predicts molecular shape based on the repulsion between electron domains (bonding pairs and lone pairs).
- **Hybridization**: The mixing of atomic orbitals to form new hybrid orbitals (e.g., sp, sp^2, sp^3) suitable for bonding.

Lattice Energy and Formal Charge

Coulomb's Law *

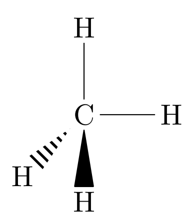
$$E_{\text{lattice}} \propto \frac{Q_1 Q_2}{d}$$

Formal Charge *

$$FC = (\text{Valence } e^-) - (\text{Non-bonding } e^-) - \frac{1}{2}(\text{Bonding } e^-)$$

Molecular Geometry Example: Tetrahedral

Centrally bonded atoms with four electron domains (e.g., CH_4) adopt a tetrahedral geometry with bond angles of approximately 109.5°.



Tetrahedral Geometry (CH_4)

Problem-Solving Skills

1. **Determining Hybridization**: Count electron domains (lone pairs + bonding regions).
   - 2 domains: sp (Linear)
   - 3 domains: sp^2 (Trigonal Planar)
   - 4 domains: sp^3 (Tetrahedral)
2. **Bond Counting**: Every single bond is 1 σ. Double bonds consist of 1 σ and 1 π. Triple bonds consist of 1 σ and 2 π.

Chapter 9: States of Matter

Core Concepts

- **Ideal Gas Law**: Relates pressure, volume, temperature, and moles. Assumes gas particles have no volume and no IMFs.
- **Kinetic Molecular Theory (KMT)**: Gas particles are in constant random motion. Temperature is a measure of average kinetic energy.
- **Intermolecular Forces (IMFs)**:
  - **London Dispersion**: Present in all molecules; strength increases with polarizability (larger electron clouds).
  - **Dipole-Dipole**: Between polar molecules.
  - **Hydrogen Bonding**: Strong dipole-dipole interaction involving H bonded to N, O, or F.

Gas Laws and Constants

Gas Laws and Constants (Provided)

Constants:

$$R = 0.0821\,\mathrm{L\,atm\,mol^{-1}\,K^{-1}}$$

$$1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ Torr}$$

Laws:

$$PV = nRT \quad \text{and} \quad \frac{P_1V_1}{n_1T_1} = \frac{P_2V_2}{n_2T_2}$$

Graham's Law of Effusion:

$$\frac{\text{rate}_1}{\text{rate}_2} = \sqrt{\frac{M_2}{M_1}}$$

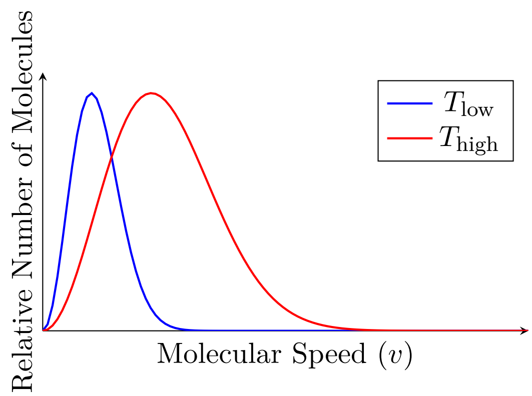
Gas Density

Gas Density *

$$d = \frac{PM}{RT}$$

Maxwell-Boltzmann Distribution

The distribution of molecular speeds changes with temperature. At higher temperatures, the peak shifts right (higher average speed) and flattens (wider distribution of speeds).



Problem-Solving Skills

1. **Gas Law Calculations**: Always convert temperature to K ($T_K = T_C + 273.15$). Ensure units match the R constant.

2. **Predicting Physical Properties**: Stronger IMFs result in higher boiling points, higher melting points, and lower vapor pressures.

Addendum: Exam Formula & Constant Reference

Provided Information

- **Temperature**: $T_{\mathrm{K}} = T_{^\circ\mathrm{C}} + 273.15$
- **Avogadro Constant**: $N_A = 6.022 \times 10^{23} 1/\mathrm{mol}$
- **Gas Constant**: $R = 0.0821\,\mathrm{L\,atm/mol\,K}$
- **Pressure**: $1\,\mathrm{atm} = 760\,\mathrm{mmHg} = 760\,\mathrm{Torr}$
- **Ideal Gas Law**: $PV = nRT$
- **Combined Gas Law**: $\frac{P_1V_1}{n_1T_1} = \frac{P_2V_2}{n_2T_2}$
- **Graham's Law**: $\frac{\mathrm{rate}_1}{\mathrm{rate}_2} = \sqrt{\frac{M_2}{M_1}}$
- **Specific Heat**: $q = mc_s\Delta T$

Information to Memorize *

- **Atomic Mass**: $\sum(\mathrm{mass} \times \mathrm{fraction\ abundance})$
- **Light**: $c = \lambda\nu$ and $E = h\nu = hc/\lambda$
- **Rydberg**: $\Delta E = -2.18 \times 10^{-18}\mathrm{J}(1/n_f^2 - 1/n_i^2)$
- **Molarity**: $M = n/V$
- **Enthalpy (Formation)**: $\sum \Delta H_f^\circ(\mathrm{prod}) - \sum \Delta H_f^\circ(\mathrm{react})$
- **Enthalpy (Bonds)**: $\sum(\mathrm{broken}) - \sum(\mathrm{formed})$
- **Lattice Energy**: $E \propto Q_1Q_2/d$
- **Solubility**: (See Table in Chapter 6)